8. Normality 가정, but 모분산 Constant σ^2

(1) a) $E(\varepsilon)=0$

$Var(\varepsilon)=\sigma^2$

b) $\sigma^2 = MSE$

c) 특정 pattern, 경향

9. a) $r^2 = 0.027 = 2.7\%$

then sales can explain 2.7% about price variation

b) Residual & lhs 관계 없음

c) is not linear function, 왜 $r^2=0.0077$ 작아짐.

(2) a) $\bar{x}=3, \bar{y}=8.4$ $\quad S_{xx} = \sum x_i^2 - n\bar{x}^2 = 10 \quad S_{xy}=15$

b: $\frac{S_{xy}}{S_{xx}} = 1.5$

$y = 1.5x + \alpha$

$\alpha = \bar{y} - 1.5\bar{x}$

$= 4.3$

c) $SSE = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 42.4 - \frac{15^2}{10}$

$= 19.9$

$MSE = \frac{19.9}{13} = 1.5307$

d) $H_0: \beta = 0$

$H_1: \beta \neq 0$

$t = \frac{\beta}{\sqrt{\frac{MSE}{S_{xx}}}} = \frac{1.5}{\sqrt{\frac{1.53}{10}}} = 3.83$

$t_{13, 0.025} = 2.160 < 3.83$, reject H_0

e) $P(t > 3.83) < 0.05$

f)

g) $x_0 = 3$

CI: $\hat{y} \pm t_{0.025,13}\sqrt{MSE\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$

$\hat{y} = 4.3 + 1.5(3)$

$= 8.8$

$8.8 \pm 2.16(0.72) = (8.11, 9.49)$

(5) b: $\frac{S_{xy}}{S_{xx}} = -0.875$

$y = -0.875x + L$

a) $x_0 = -1$

$\hat{y} = 2 - 0.875(-1) = 2.875$

94% CI: $\hat{y} \pm t_{0.025,13}\sqrt{MSE\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$

$= (2.01, 3.74)$

b) $x_0 = 1$

$\hat{y} = 1.125$

94% CI $= 1.125 \pm 3.441\sqrt{0.0773\left(1+\frac{1}{3}+\frac{1}{8}\right)}$

$= 1.125 \pm 1.975$

$= (-0.79, 3.02)$

c) when $x_0 = \bar{x}$ 시, position = min

13) alpha

b) $b_0 = 90.88$, $b_1 = 270.98$

$\hat{y} = 90.88 + 270.98x$

c) $H_0: \beta_1 = 0$

$H_a: \beta_1 \neq 0$

$p = 0.001 < 0.05$ reject H_0

d) $x_0 = 0.20$

$\hat{y} = 90.88 + 270.98 \times 0.20 = 145.08$

90% CI: $145 \pm t^*_{8,0.05} \sqrt{MSE\left(\frac{1}{10} + \frac{(0.20-\bar{x})^2}{S_{xx}}\right)}$

$= (112.70, 177.46)$

14) a) $H_0: \rho = 0$

$H_a: \rho > 0$

$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = 0.929$

$t = r\sqrt{\frac{n-2}{1-r^2}} = 0.92$

90% $\Rightarrow t_{0.05,9} = 1.833 > 0.92$, non reject H_0

b) 회귀분석을 통한 hypothesis test

correlation analysis

b) $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = 0.919$

15) a) $S_{xx} = 12.59$

$S_{yy} = 410.69$

$S_{xy} = 11.73$

$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = 0.174$

b) $H_0: \rho = 0$ $H_a: \rho \neq 0$

$t = r\sqrt{\frac{n-2}{1-r^2}} = 0.58$

$t_{0.025,10} = 2.228 > 0.56$

$\therefore$ non-reject H_0

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